

ID:

Name:

Brac University

Semester: Fall 2025

Course Code: CSE250

Circuits and Electronics

Set

A

Assessment: *Midterm*

Duration: 1 hour 20 minutes

Date: December 01, 2025

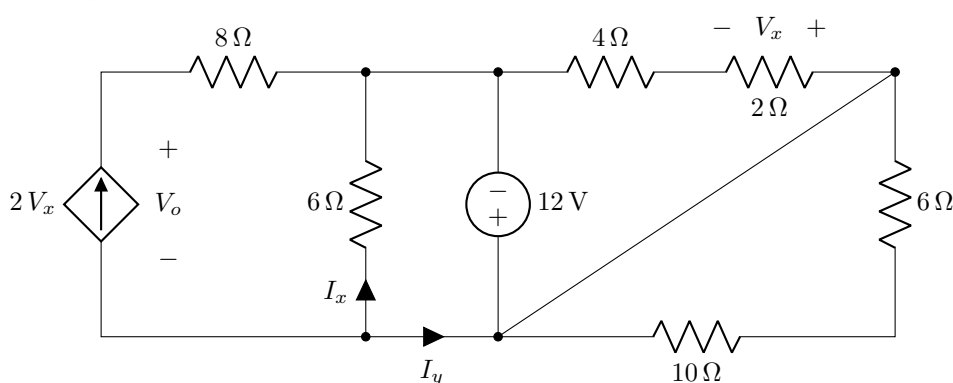
Full Marks: 55

- ✓ Phones must be turned off. Using/carrying any notes during the exam is not allowed.
- ✓ At the end of the exam, both the **answer script** and the **question paper** must be returned to the invigilator.
- ✓ **All 4 question(s) are compulsory.** Marks allotted for each question are mentioned beside each question.
- ✓ All calculated values must be accompanied by appropriate units.
- ✓ Symbols have their usual meanings.

■ Question 1 of 4

[CO1] [8 marks]

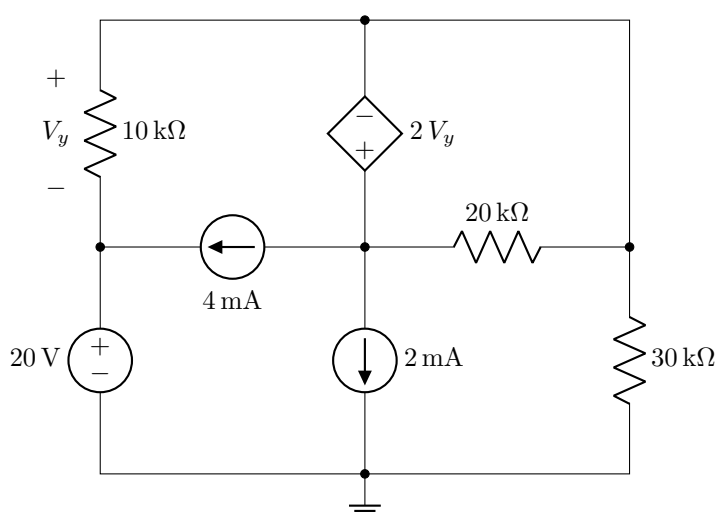
For the circuit shown below,

Apply the **Circuit Laws** to answer the following questions–

- (a) [5 marks] If the power supplied by the dependent source is 416 W, determine V_o .
- (b) [3 marks] Determine the currents I_x and I_y .

■ Question 2 of 4

[CO3] [25 marks]



For the circuit shown above,

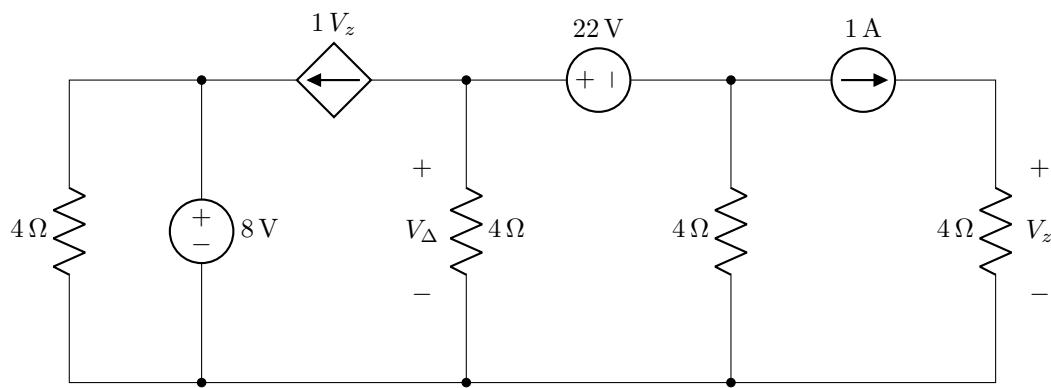
- (a) [17 marks] **Apply** either **Nodal Analysis** or **Mesh Analysis** to determine the power of (i) the 2mA current source and (ii) the 20 V voltage source with the appropriate $\pm$ sign and units.^{††} Also, mention in each case whether the source is supplying or absorbing power.
- (b) [8 marks] **Apply** the alternative method that you did not use in (a) to formulate all the equations needed to solve the circuit. You do not need to simplify or solve the equations.

^{††}Node voltage or mesh current variables must be labeled on the diagram

■ Question 3 of 4

[CO2] [12 marks]

Apply the **Principle of Superposition** in the following circuit to determine the voltage V_{Δ} .



■ Question 4 of 4

[CO3] [10 marks]

The monkey, holding onto a rope from the roof of the following house of resistors, may let go of the rope at any time and land on the floor. When it does, determine the equivalent resistance it sees between its initial position and its final position.
(Hint: redraw the circuit without the resistors that are shorted.)

